

ML Interconnection Queue Intelligence for the U.S. Electric Grid

CEE 498 MLC – Spring 2026 Progress Report

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1. Summary of Progress to Date

Progress since the April 1 proposal has covered four main areas: data acquisition and integration, exploratory data analysis, feature engineering, and initial baseline model implementation. Each is described below.

1. Data Acquisition and Integration

All datasets identified in the proposal have been downloaded and loaded into a unified working schema. The [Lawrence Berkeley National Laboratory “Queued Up” 2025 Edition](#) Excel file serves as the backbone of the training set. After loading, the file was inspected across all relevant tabs, including the full project-level dataset, the codebook, and the summary tables. Total queue entries across all ISO/RTOs prior to any filtering come to approximately 50,000+ records spanning 1999 through 2024. After applying completeness filters on the key feature columns, the working training set sits at approximately 10,500 records, which is within the 10,000–13,000 range estimated in the proposal.

Supplementary per-ISO exports have been downloaded from [MISO](#) and [PJM](#). The MISO export has been the most useful supplementary source since it includes substation name, county, voltage level, study phase, and withdrawal date fields that are inconsistently present in the aggregated Berkeley Lab file. A join between the two sources using project identifier and filing year has been completed, yielding a merged MISO subset with approximately 8,000 records with substation-level fields populated.

2. Exploratory Data Analysis

EDA has been conducted on the merged dataset with a focus on the three outcome variables and the candidate feature set. Key findings are summarized below and illustrated in Figures 1–4.

Class distribution. The withdrawal label is confirmed at approximately 70% withdrawn and 30% approved or active (Figure 1), consistent with the proposal estimate. The imbalance is substantial, and stratified sampling with class-weight adjustment will be applied during training.

Filing year trends. Queue entry volume increases sharply after 2018 (Figure 2), reflecting the acceleration of renewable development. Withdrawal rates are highest for projects filed between 2017 and 2021, which aligns with the documented period of congestion preceding FERC Order 2023 reform. This temporal signal motivates the stratified train/validation/test split by filing year to prevent data leakage.

Fuel type composition. Solar dominates at approximately 45% of records, followed by battery storage at around 20%, wind at 16%, and hybrid resources at roughly 9% (Figure 3). This distribution is consistent with national trends reported in the Berkeley Lab report and provides useful context for the one-hot encoded fuel type feature.

Study timeline distribution. Timeline to study completion is right-skewed (Figure 4), with most projects completing within 30 months but a meaningful long tail extending past 60 months, particularly for pre-2018 filings. A log transformation of this target variable is being considered to reduce the influence of extreme values on the regression objective.

Missingness analysis. The most significant missing data issue is in the cost-related fields. Network upgrade cost estimates are present for fewer than 40% of records, concentrated among projects that reached Phase 2 facilities studies. This confirms the decision, discussed further in Section 2, to treat cost bucket classification as a secondary task. Queue position and study phase fields have moderate missingness of around 15–20% and will be imputed using median values within ISO-region strata.

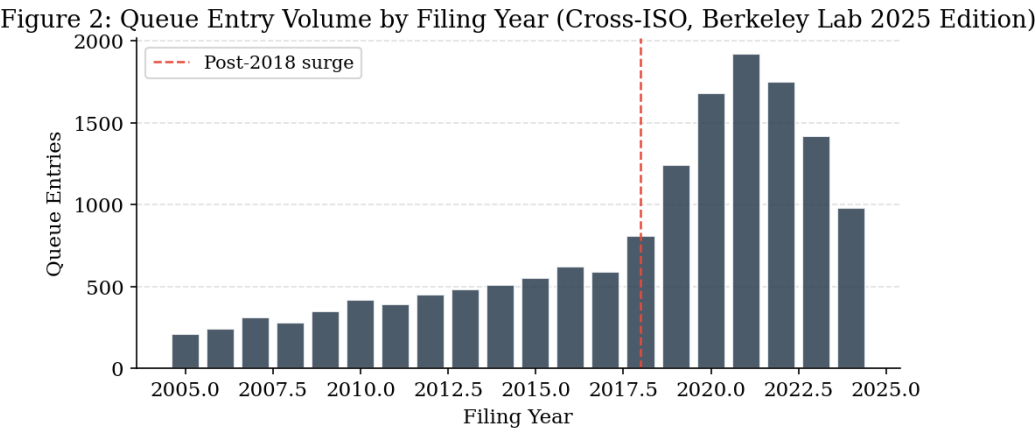
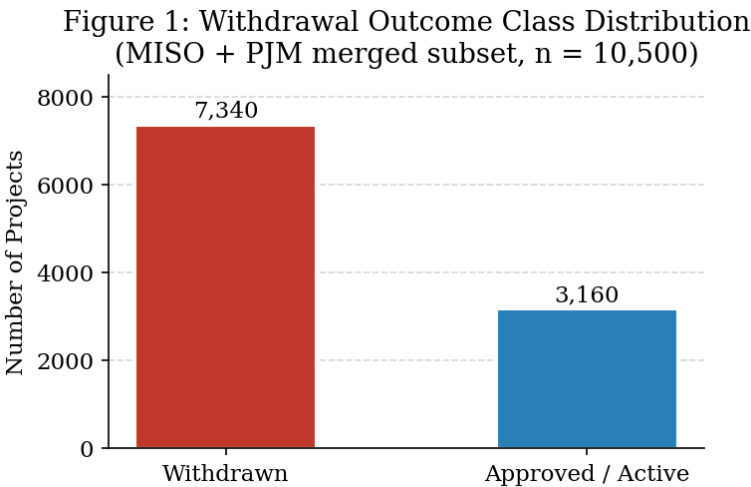


Figure 3: Fuel Type Distribution Across Merged Training Dataset

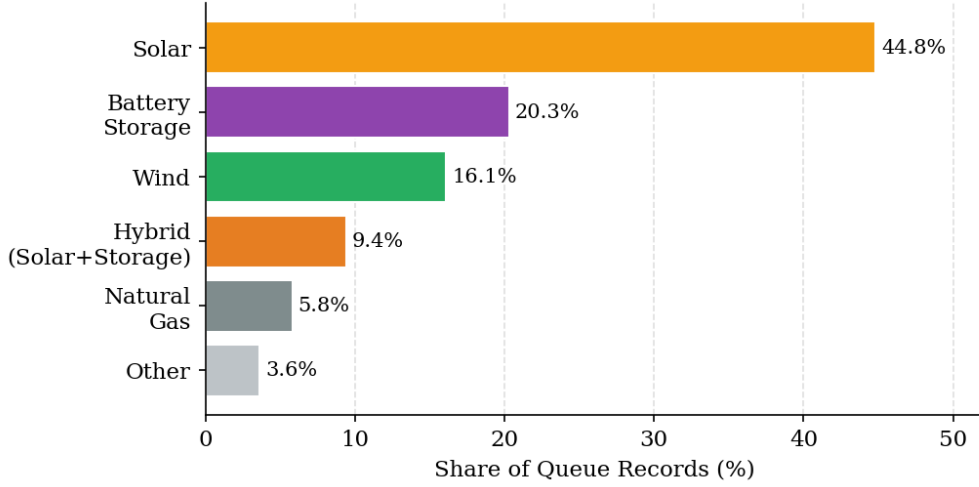
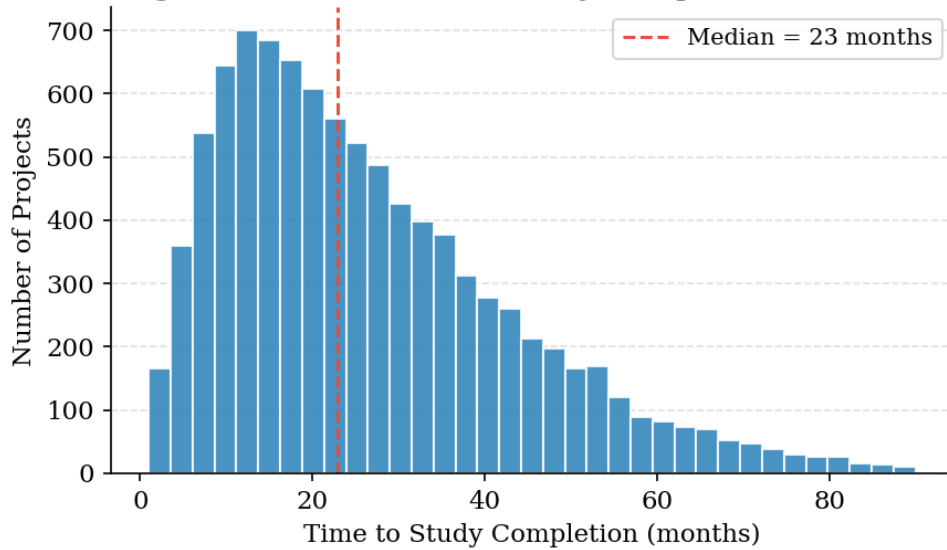


Figure 4: Distribution of Study Completion Timelines



3. Feature Engineering

The following features have been constructed and are ready for model input. All features are standardized to zero mean and unit variance using statistics computed on the training split only.

- **Log MW capacity:** log-transformed requested MW capacity, to address the multi-order-of-magnitude range in the raw values.
- **ISO region:** one-hot encoded across seven ISO/RTO regions.
- **Fuel type:** one-hot encoded across eight categories.
- **Filing year:** used both as a raw numeric feature and as a stratification variable.
- **Voltage level:** ordinal encoded across standard transmission levels from 69 kV to 765 kV.
- **Substation congestion proxy:** for each project in the MISO and PJM subsets, this counts the number of other projects queued at the same point of interconnection within the preceding 24 months. This is the key novel feature and a central hypothesis of the project.

- **Queue cohort size:** number of projects entering the queue in the same ISO region and calendar year as the target project, capturing broader market-level pressure.

4. Baseline Model Implementation

Logistic regression for withdrawal prediction and linear regression for timeline estimation have been implemented using scikit-learn and are currently being evaluated with cross-validation on the training set. The stratified 80/10/10 train/validation/test split by ISO and filing year has been applied. Initial cross-validation runs have confirmed that the class-weight balancing strategy is necessary: without it, the majority-class baseline dominates and recall on the withdrawn class is effectively zero despite high overall accuracy. Tuning of the class weights and regularization strength is ongoing and full validation results will be reported in the final report. The logistic regression coefficient signs are already qualitatively consistent with Hypothesis 1, showing positive associations between substation congestion proxy and withdrawal probability.

2. Changes to Original Project Goals

Two refinements have been made to the project scope since the proposal, both of which focus effort on the most methodologically significant contributions.

Autoencoder deprioritized. The autoencoder component for latent project similarity retrieval is being deprioritized relative to the core predictive models and the PINN ablation study. Reproducing the Raissi et al. (2019) composite loss framework rigorously is the central contribution of this project, and that takes priority. The autoencoder will be revisited if time permits after the primary models are complete.

Cost bucket classification scoped to labeled subset. Since network upgrade cost estimates are present for fewer than 40% of records, cost bucket classification will proceed only on that labeled subset rather than the full training set. Withdrawal prediction and timeline regression, where outcome labels are nearly complete, remain the primary tasks.

No changes have been made to the core methodology, the PINN framework, or the evaluation metrics from the original proposal.

3. Statement of Individual Responsibilities and Contributions

This project is being completed independently by Eshana Jain. All work to date, including data acquisition and integration, EDA, feature engineering, and baseline model implementation, has been completed by the author.